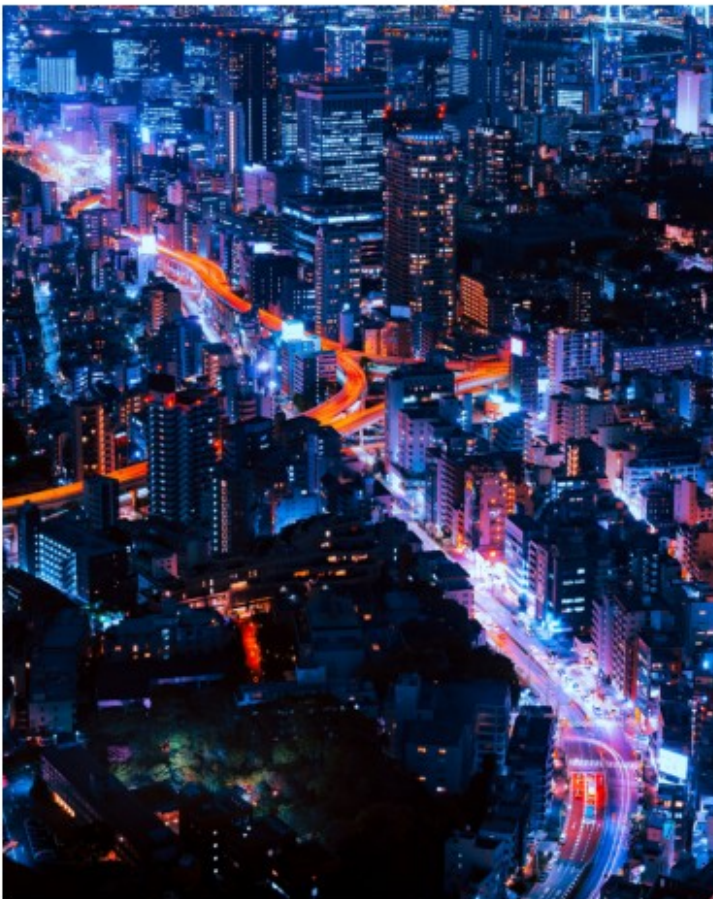


```
import cv2
import numpy as np
import matplotlib.pyplot as plt

# Load an image
image = cv2.imread('input_image.jpg')
if image is not None:
    # Convert BGR to RGB for correct plotting with matplotlib
    image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
    plt.figure(figsize=(6, 6))
    plt.imshow(image_rgb)
    plt.title('Loaded Image')
    plt.axis('off')
    plt.show()

    # Convert the image to grayscale for texture analysis
    gray_image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
```

Loaded Image



```
# Feature extraction
color_features = image.reshape((-1, 3)) # Reshape the color image into
```

a 2D array

```
texture_features = gray_image.reshape(-1) # Reshape the grayscale image into a 1D array
```

```
# Plot color histogram
```

```
plt.figure(figsize=(12, 6))
```

```
plt.subplot(121)
```

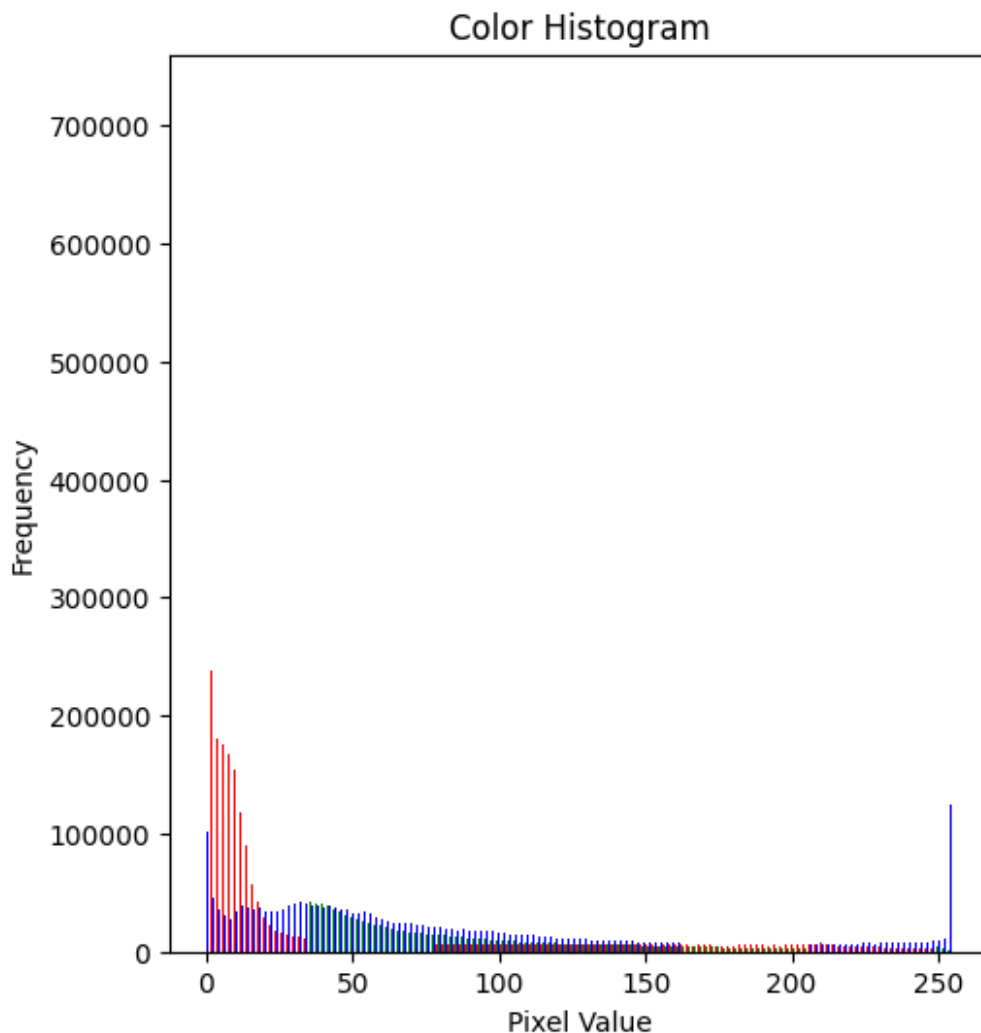
```
plt.hist(color_features, bins=256, color=['blue', 'green', 'red'])
```

```
plt.title('Color Histogram')
```

```
plt.xlabel('Pixel Value')
```

```
plt.ylabel('Frequency')
```

```
Text(0, 0.5, 'Frequency')
```



```
if image is not None:  
    # Plot texture histogram
```

```
plt.subplot(122)
plt.hist(texture_features, bins=256, color='gray')
plt.title('Texture Histogram')
plt.xlabel('Pixel Value')
plt.ylabel('Frequency')

plt.show()
else:
    print('Image could not be found or could not be loaded.')
```

